Ishaan Karan

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Education

International Institute of Information Technology — Hyderabad, India
Bachelor of Technology in Computer Science, Master of Science in Computational Linguistics — *Aug 2023 – May 2028*

Relevant Coursework: Data Structures & Algorithms, Design and Analysis of Algorithms, Operating Systems & Networks, Advanced NLP, Data & Applications, Design & Analysis of Software Systems, Responsible and Safe AI Systems, Applications of Language Models

Whitmore High School — London, United Kingdom
A Levels in Further Mathematics- A, Mathematics-A*, Physics-A, Chemistry-A;* — *Sep. 2021 – Jun 2023*

Experience

Undergraduate Researcher — May 2025 – Present
Language Technologies Research Centre
- Building deep learning models for long-document understanding under Dr. Rajakrishnan Rajkumar. Research focuses on saliency detection, summarization, and span extraction in narratives and social media text.

Software Engineer Intern — Jan 2025 – April 2025
Automaxis
- Built a decentralized system for Certificates of Origin using Algorand NFTs, Pinata IPFS, Node.js APIs, and MongoDB with real-time tracking and role-based workflows.

Publications

Span-Consistency Network for Conspiracy Marker Extraction — SemEval @ ACL 2026
- Designed a span extraction architecture with Cross-Marker Attention to model inter-label dependencies and a custom regularization loss to prevent prediction collapse, earning 2nd place in SemEval-2026 Task 10.

BAM-PQ: Bloom-Aligned Matryoshka Embeddings with Per-Query Routing — Under Review, ARR
- Introduced per-query dimension routing for dense retrieval using Bloom's taxonomy; outperforms fixed-prefix Matryoshka baselines by 3-6 R@10 points at 25-43% of embedding dimensions across five encoder backbones.

Projects

The Cross-Modal Safety Gap | *Python, PyTorch, Sparse Autoencoders* | GitHub — Jan 2026 – April 2026
- Diagnosed why aligned multimodal LLMs refuse harmful text but comply with images or audio; trained sparse autoencoders on two 7B models to isolate ∼50 safety features (0.3% of features) that fire on harmful text but never on non-text inputs.
- Localized the safety circuit to layers 12–13 via three independent methods and showed text and non-text representations are near-orthogonal, bounding inference-time defenses.

Distributed Document Collaboration System | *C, TCP Sockets, POSIX* | GitHub — Nov 2025
- Built a distributed, multi-user document system with a central name server coordinating multiple storage servers and concurrent clients, routing requests and brokering direct client–server connections for high-throughput reads, writes, and streaming.
- Implemented sentence-level locking for concurrent multi-user editing, owner-controlled read/write access control, durable persistence of files and metadata, and fast file lookup through indexed search with LRU caching.

Sparse Mixture-of-Experts for Extreme Summarization | *Python, PyTorch* | GitHub — Nov 2025
- Built a sparse Mixture-of-Experts Transformer from scratch with Grouped Query Attention, two routing strategies (hash and top-k token-choice), and a load balancer with an auxiliary balancing loss.
- Implemented LoRA-based experts from scratch, trained on XSum for extreme summarization, and benchmarked against BART, a fine-tuned T5, and an instruction-tuned LLaMA using ROUGE, BERTScore, and factuality metrics.

Custom C Shell | *C, POSIX* | GitHub — Aug 2025
- Built a POSIX-compliant Unix shell in C with a modular, multi-file architecture, including a recursive-descent parser that validates commands against a context-free grammar and built-in commands for directory navigation, file listing, and command history persisted across sessions.
- Implemented external command execution, input/output redirection, and multi-stage command pipelines, with sequential and background execution and full job control, including foreground/background switching and signal-based process management.

Positions of Responsibility

Sports Captain | *IIIT Hyderabad* — Aug 2025 – May 2026
- Led Night Cricket, IIIT-H's largest sporting event with 1000+ participants and 120+ teams; orchestrated end-to-end operations including venue management, fixture scheduling, and live coordination.

Achievements and Awards

British Mathematical Olympiad (BMO1) - Merit Award | **NTSE Stage 1** - State Rank 16 | **IOQM** - Top 300 in state

Technical Skills

Languages: Python, C/C++, JavaScript, SQL, Bash
Tools & Frameworks: PyTorch, Hugging Face Transformers, PEFT/LoRA, CUDA, Weights & Biases, SHAP, scikit-learn, NumPy, Pandas, Matplotlib, spaCy, NLTK, React.js, Node.js, MongoDB, Git, Linux, POSIX, TCP Sockets